In [35]:

```python
# Importando las Librerias
import pandas as pd
import matplotlib.pyplot as plt
from wordcloud import WordCloud, STOPWORDS
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import linear_kernel
from sklearn.feature_extraction import text
from request.utils import dwelling_list
import json

dict_data = dwelling_list(100)['results']

# Serializing json
json_object = json.dumps(dict_data)

# Getting data from Turismo Real Backend
df = pd.read_json(json_object)
```

In [36]:

```python
df.head()
```

Out[36]:

	id	rooms	products
0	1	LIVING DORMITORIO DORMITORIO	Televisor Juguera Cama 2 plazas Televisor Escr...
1	2	LIVING DORMITORIO DORMITORIO COCINA	Televisor Juguera Cama 2 plazas Televisor Escr...
2	3	LIVING DORMITORIO BALCÓN	Televisor Juguera Cama 2 plazas Televisor Escr...
3	4	LIVING DORMITORIO DORMITORIO	Televisor Juguera Cama 2 plazas Televisor Escr...
4	5	LIVING DORMITORIO	Televisor Juguera Cama 2 plazas Televisor Escr...

In [37]:

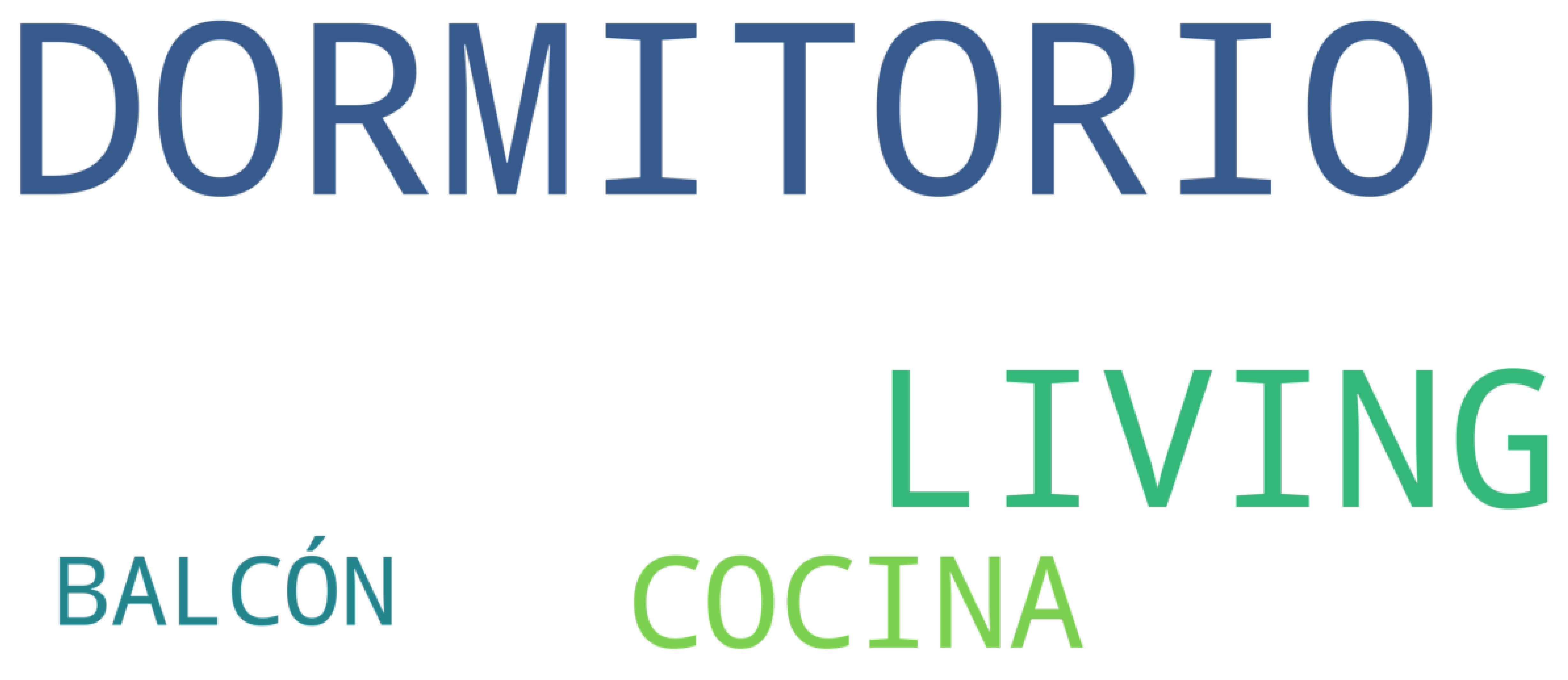
```python
## ANALISIS
df['rooms'] = df['rooms'].astype('str')
df['products'] = df['products'].astype('str')
```

In [38]:

```python
rooms_corpus = ' '.join(df['rooms'])
products_corpus = ' '.join(df['products'])
```

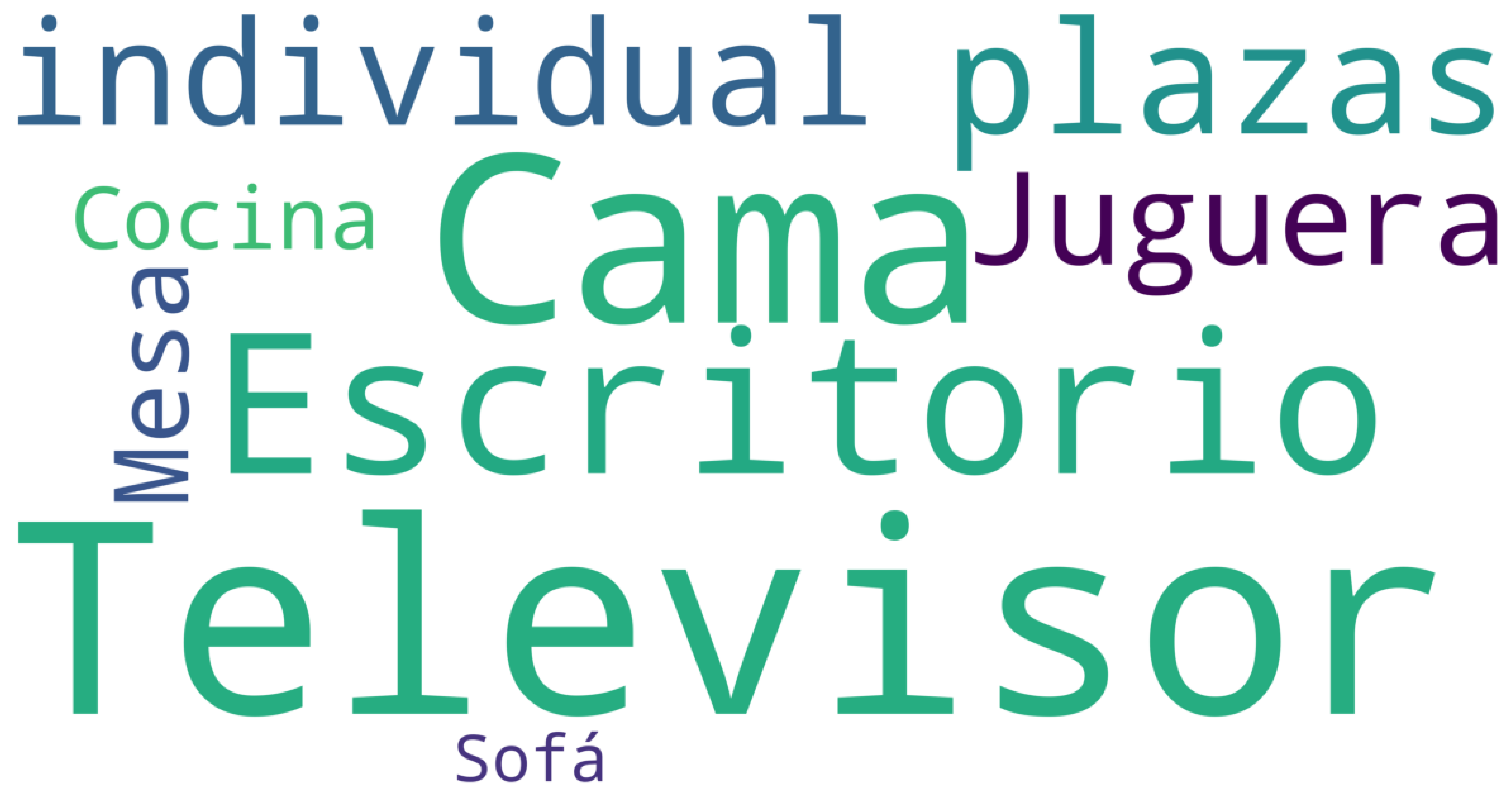
In [39]:

```python
name_wordcloud = WordCloud(stopwords = STOPWORDS, background_color = 'white', height = 2000, width = 4000).generate(rooms_corpus)
plt.figure(figsize = (16,8))
plt.imshow(name_wordcloud)
plt.axis('off')
plt.show()
```



In [40]:

```python
name_wordcloud = WordCloud(stopwords = STOPWORDS, background_color = 'white', height = 2000, width = 4000).generate(products_corpus)
plt.figure(figsize = (16,8))
plt.imshow(name_wordcloud)
plt.axis('off')
plt.show()
```



In [41]:

```python
# Creamos una columna que contiene la combinacion de columnas productos y salas
df['content'] = df[['rooms', 'products']].astype(str).apply(lambda x: ' // '.join(x), axis = 1)
```

In [42]:

```python
# Fillna
df['content'].fillna('Null', inplace = True)
```

In [43]:

```python
# Entrenando el recomendador
my_stop_words = list(text.ENGLISH_STOP_WORDS.union(["spanish"]))
tf = TfidfVectorizer(analyzer = 'word', ngram_range = (1, 2), min_df = 0, stop_words = my_stop_words)
tfidf_matrix = tf.fit_transform(df['content'])
```

In [44]:

```python
# Similitud de coseno
cosine_similarities = linear_kernel(tfidf_matrix, tfidf_matrix)
```

In [45]:

```python
results = {}
for idx, row in df.iterrows():
    similar_indices = cosine_similarities[idx].argsort()[:-100:-1]
    similar_items = [(cosine_similarities[idx][i], df['id'][i]) for i in similar_indices]
    results[row['id']] = similar_items[1:]
```

In [46]:

```python
# Prediction
def item(id):
    name   = df.loc[df['id'] == id]['content'].tolist()[0].split(' // ')[0]
    desc   = ' \nProducts: ' + df.loc[df['id'] == id]['content'].tolist()[0].split(' // ')[1][0:165] + '...'
    prediction = name  + desc
    return prediction

def recommend(item_id, num) -> list:
    recommendations = []
    print('Recommending ' + str(num) + ' dwellings similar to DWELLING ID :' + str(item_id))
    print('---')
    recs = results[item_id][:num]
    for rec in recs:
        print('\nRecommended: Dwelling ' + str(rec[1]) + ' ' + item(rec[1]) + '\n(score:' + str(rec[0]) + ')')
        recommendations.append({'id' : rec[1] , 'score' : round(rec[0],2)})
    return recommendations
```

In [47]:

```python
recommend(item_id = 1, num = 3)
```

```
Recommending 3 dwellings similar to DWELLING ID :1
---

Recommended: Dwelling 1 LIVING DORMITORIO DORMITORIO
Products: Televisor Juguera Cama 2 plazas Televisor Escritorio Cama individual Televisor Escritorio Mesa ...
(score:1.0)

Recommended: Dwelling 5 LIVING DORMITORIO
Products: Televisor Juguera Cama 2 plazas Televisor Escritorio Cama individual...
(score:0.8729786852585604)

Recommended: Dwelling 2 LIVING DORMITORIO DORMITORIO COCINA
Products: Televisor Juguera Cama 2 plazas Televisor Escritorio Cama individual Televisor Escritorio Mesa  Cocina...
(score:0.7806061081798649)
```

Out[47]:

```
[{'id': 1, 'score': 1.0}, {'id': 5, 'score': 0.87}, {'id': 2, 'score': 0.78}]
```